Dural Venous Sinus Stenting in Pediatric Refractory Idiopathic Intracranial Hypertension: A Case Series and Literature Review

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■ BACKGROUND: **Management of pediatric idiopathic intracranial hypertension (IIH) presents significant challenges. Although venous sinus stenting has emerged as an effective minimally invasive treatment for adult IIH, its efficacy and safety in the pediatric population is not well studied.**

■ METHODS: **Three cases of pediatric IIH refractory to medical management and treated with venous sinus stenting were retrospectively reviewed.**

■ RESULTS: **The first case, a 17-year-old girl with a body mass index (BMI) of 57.06 kg/m^2, cerebrospinal fluid (CSF) opening pressure of 38 cm H_2O, and a pressure gradient of 15 mm Hg across the venous sinus stenosis, presented with bilateral optic disc edema, visual deficits, and a worsening 3-day headache. The second case, a 14-year-old girl with a BMI of 57.06 kg/m^2, CSF opening pressure of 28 cm H_2O, and a pressure gradient of 12 mm Hg across the venous sinus stenosis presented with bilateral optic disc edema, visual deficits, nausea, and headache. The last case, a 12-year-old girl with a BMI of 30.47 kg/m^2, opening pressure of 39.5 cm H_2O, and a pressure gradient of 7 mm Hg across the venous sinus stenosis, presented with diplopia, headache, and ataxia. All cases had been managed medically with acetazolamide before venous stenting. All 3 patients demonstrated symptomatic and visual improvement at 6-month follow-up, and there were no major surgical complications noted in these patients.**

■ CONCLUSIONS: **Although IIH is difficult to manage in the pediatric population, we report successful implementation of venous sinus stenting in the management of pediatric IIH. There were no major operative complications.**

INTRODUCTION

Normal increased intracranial pressure is maintained at a constant level by self-regulating processes that adapt to alterations in cerebrospinal fluid (CSF) production and absorption rates.[1] Pediatric idiopathic intracranial hypertension (PIIH) is a disorder of unknown etiology characterized by increased intracranial pressure (ICP) without structural lesions in the brain.[2] The incidence of IIH is 0.49—0.63 per 100,000 in the pediatric population, appearing at an increased rate in Black children.[3] Typical patient symptoms in both adults and pediatrics include headache, pulse synchronous tinnitus, diplopia, and papilledema, which if left untreated may result in significant visual impairment. Vision loss is one of the most devastating complications of idiopathic intracranial hypertension (IIH), with a rate of bilateral blindness reaching 10%.[4-6] Nonsurgical treatment modalities are aimed at reducing production of CSF primarily through reduction of body weight and diuretics (most commonly acetazolamide).[7] If IIH is refractory to medical treatment, surgical options include CSF shunting, serial lumbar punctures to divert CSF, or optic nerve fenestration.[8] Venoplasty has been proposed as a novel minimal invasive approach for this

Key words
- Anticoagulation
- Hemorrhage
- Idiopathic intracranial hypertension
- Papilledema
- Pediatrics
- Venous sinus stent

Abbreviations and Acronyms
BMI: Body mass index
CSF: Cerebrospinal fluid
IIH: Idiopathic intracranial hypertension
MRI: Magnetic resonance imaging
MRV: Magnetic resonance venography
ONSD: Optic nerve sheath decompression
PIIH: Pediatric idiopathic intracranial hypertension

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condition. This procedure implements a stent placement that reduces luminal stenosis and lowers the pre- to poststenotic venous pressure gradient and increases the pressure gradient across the arachnoid villi to enhance CSF reabsorption and thereby reduce the volume of CSF in the ventricles, leading to a reduction in intracranial pressure and improvement in symptoms.[9] In the pediatric population, traditional CSF diversion procedures and lumbar punctures are associated with increased morbidity, limiting the efficacy of surgical approaches.[9,10] Data on the effectiveness of venous sinus stenting in patients with pediatric IIH are extremely limited, with only 36 cases reported in the English literature to date.[11-21] The use of venoplasty as a surgical treatment modality in the pediatric population for IIH could prove an invaluable resource to mitigate symptoms refractory to medical management. Here we illustrate utilization of venous sinus stenting in the management of PIIH and review the current literature to highlight outcomes, efficacy, and complications associated with venous stenting in PIIH.

METHODS

Three cases of pediatric IIH refractory to conventional medical management who were managed with venous sinus stenting were retrospectively reviewed. All patients had standard workup including magnetic resonance venography (MRV), catheter angiography, venous manometry, visual assessment, and lumbar puncture confirming stenosis and elevated intracranial pressure. Postoperative outcomes included visual acuity, papilledema, symptoms resolution, procedural complications, postoperative medical management, and long-term complications.

CASE DESCRIPTIONS

Case 1

A 17-year-old girl with a body mass index (BMI) of 57.06 kg/m^2, history of seizure disorder, obstructive sleep apnea, and bilateral optic disc edema presented with a 3-day headache localizing to the parietal region and worsening with movement with associated fatigue. The patient's mother also noted medial eye deviation. At 13 years of age, the patient had acutely presented with headache, and fundoscopic examination demonstrated bilateral optic disc edema. She subsequently was diagnosed with PIIH when lumbar puncture revealed opening pressure of 28 cm H_2O and magnetic resonance imaging (MRI) and MRV were unrevealing. Aside from a slight prominence of fluid signal about the optic nerve sheaths, the MRV revealed normal venographic findings. The patient was then managed via weight loss and acetazolamide (up to a maximum dose of 4000 mg/d).

Despite medical management, the patient had recurrent symptoms with worsening papilledema. A lumbar puncture was completed, and opening pressure was measured at 38 cm H_2O. MR venography revealed focal narrowing of the lateral aspect of the transverse sinuses bilaterally. Dural venous sinus manometry confirmed bilateral narrowing of the lateral transverse sinuses and sigmoid sinuses and demonstrated hemodynamically significant stenosis with a pressure gradient of 11 mm Hg.

The patient was successfully treated with venous sinus stenting and initially reported clinical improvement and fundoscopy demonstrated reduction in papilledema. However, shortly after stent placement, the patient experienced a significant deterioration in psychiatric condition, which led to overeating and a subsequent uncontrolled increase in BMI. Nine months later, the patient represented with worsening headaches, and optic nerve edema. Dural venous sinus manometry revealed severe stenosis of the transverse sinus proximal to the previously placed stents with a trans-stenotic pressure gradient of 22 mm Hg (Figure 1A). An additional 10- × 24-mm stent was deployed in the region of the right transverse sinus stenosis. Manometry revealed a reduction in torcula-transverse sinus pressure gradient (Figure 1B). The patient was placed on dual antiplatelet therapy (low dose aspirin and clopidogrel), and acetazolamide (4000 mg/d) was continued.

Preoperatively, her visual acuity without correction was 20/25 bilaterally, with 3+ bilateral disc edema. After the first stent

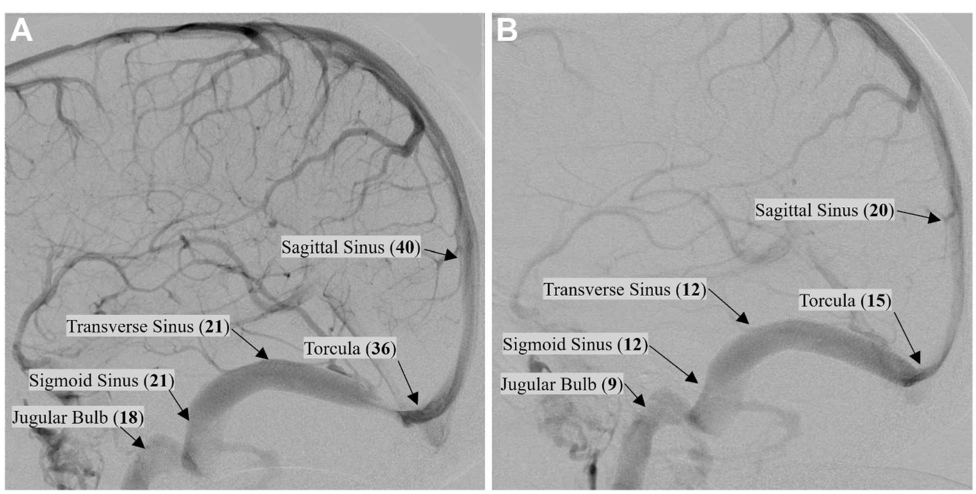

Figure 1. (**A**) Venous sinus venography/manometry demonstrating severe stenosis of the transverse sinus proximal to the previous stent. (**B**) Venous sinus venography/manometry demonstrating resolution of torcula-transverse sinus pressure. All manometry values are in millimeters of mercury.

placement, her vision was 20/30 (right eye) and 20/25 (left eye), with resolution of disc edema. After the second stent placement, her visual acuity without correction was 20/50 (right eye) and 20/40 (left eye), with grade 1 disc edema. At the last ophthalmology follow-up (32 months), visual acuity without correction was 20/30 (right eye) and 20/40 (left eye), with mild disc gliosis.

After stenting, antiplatelet therapy (aspirin 81 mg/d and clopidogrel 75 mg/d) was planned for 3 months, after which low-dose aspirin would be continued. However, the patient experienced recurrent epistaxis and oral bleeding, approximately once a day. This was particularly challenging to manage because of her continuous positive airway pressure use for obstructive sleep apnea, leading to low compliance with antiplatelet therapy. At the 10-month radiographic follow-up, the right transverse/sigmoid sinus stents were patent, and venous manometry revealed a torcula-sigmoid sinus pressure of 2 mm Hg.

Case 2

A 14-year-old girl with obesity of BMI of 34.2 kg/m^2 presented with worsening bilateral papilledema, blurred vision, nausea, and worsening bifrontal headache. Three months prior, the patient experienced significant worsening of vision, described as "stars" on the right eye and a "black spot" on the left. At this point, a lumbar puncture revealed opening pressure of 28 cm H_2O. MRI revealed bilateral narrowing of the distal transverse sinuses. Neuro-ophthalmology managed the patient with acetazolamide (4000 mg/d) for PIIH; however, despite increasing dose, the patient experienced worsening of symptoms. The patient underwent catheter angiography and sinus manometry was completed, redemonstrating moderate stenosis at the left transverse sigmoid junction with a transverse-sigmoid sinus pressure gradient of 12 mm Hg (Figure 2). The patient underwent venous angioplasty using a Wall stent (10 × 31 mm). After venoplasty, venous manometry revealed a substantial reduction in the pressure gradient to <2 mm Hg.

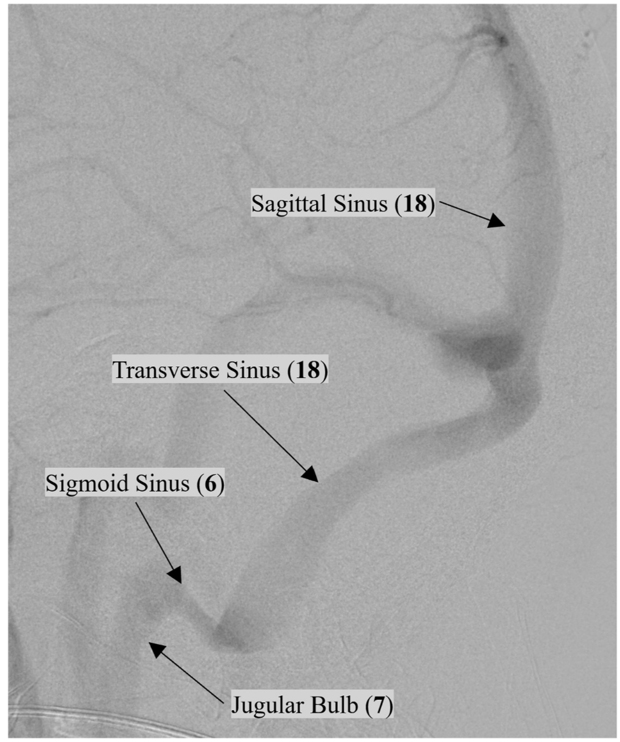

Figure 2. Venous sinus venography/manometry demonstrating moderate intrinsic stenosis at the left transverse sigmoid junction with a transverse-sigmoid sinus pressure of 12 mm Hg. All manometry values are in millimeters of mercury.

Preoperatively, the patient's visual acuity without correction was 20/100 (right eye) and 20/400 (left eye), with 5+ bilateral disc edema. After stent placement, vision was 20/50 (right eye) and 20/150 (left eye), with a 3- to 4-disc edema. Neuro-ophthalmology follow-up at 40 months revealed visual acuity without correction of 20/20 (right eye) and 20/25 (left eye), with mild disc pallor and arteriolar sclerosis.

The patient was placed on dual antiplatelet therapy (81 mg/d aspirin and 75 mg/d clopidogrel), and the dose of acetazolamide was gradually tapered (39 months after stent placement the patient stopped taking acetazolamide). One month after initiation of antiplatelet therapy, the patient experienced excessive menstrual bleeding, requiring temporary cessation of antiplatelet therapy. After stabilization of anemia, the patient was placed on low-dose aspirin (81 mg/d). At the 17-month follow-up in clinic, the patient recovered well and noted improvement in headaches and improved vision (at which point acetazolamide dose was 750 mg/d).

Case 3

A 12-year-old girl with a BMI of 30.47 kg/m^2 and a history of IIH presented with blurry vision, headache, and ataxia. The patient's PIIH had been previously diagnosed via an increased opening pressure and MRI demonstrating focal stenosis of the right transverse sinus. The patient was managed with multiple lumbar punctures and acetazolamide. MRI demonstrated focal stenosis of the right transverse sinus. The patient's diplopia resolved with medical management.

However, because of worsening ataxia, venography and transverse sinus manometry were completed, redemonstrating moderate stenosis at the left transverse sinus with a transverse sinus pressure of 7 mm Hg (Figure 3). A venous stent (8 × 29 mm) was then placed at the site of stenosis, after which venous phase imaging demonstrated no significant stenosis. Postoperatively, the patient was placed on dual antiplatelet therapy (aspirin 81 mg/d and clopidogrel 75 mg/d) and continued acetazolamide (4000 mg/d). The patient recovered well, and there were no surgical complications.

Within 1 week of initiating antiplatelet therapy, the patient experienced severe, recurrent epistaxis. Dual antiplatelet therapy was continued for 3 months. Because of continued hemorrhagic events, clopidogrel was eliminated from the antiplatelet therapy, and low-dose aspirin was continued at last follow-up. Angiographic follow-up at 10 months demonstrated patent stent with improved pressure gradient. On follow-up in clinic at 18 months, the patient demonstrated resolution of all preoperative symptoms and had been tapered down to a lower dose of acetazolamide.

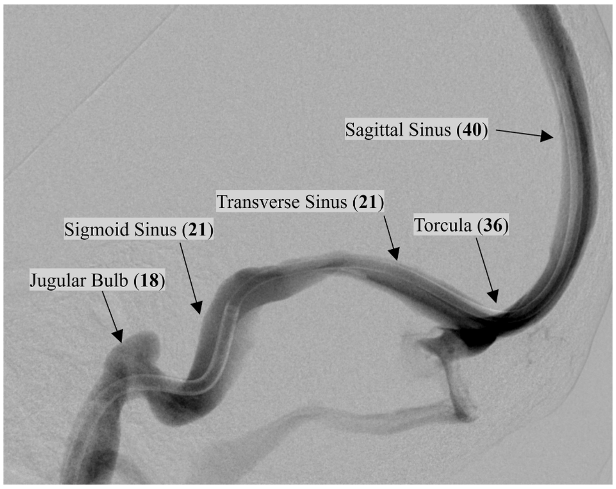

Figure 3. Venous phase angiography/manometry demonstrating focal stenosis of the right transverse sinus and a transverse sinus pressure of 7 mm Hg. All manometry values are in millimeters of mercury.

DISCUSSION

A literature review of PIIH was completed through PubMed, Cochrane Library, and Embase databases until May 2024 for reports of PIIH managed with dural venous sinus stenting. The results of this review yielded 36 cases of PIIH.[11-21]

Demographics

The average age ± SD of the patients was 12.28 ± 4.74 years, with the youngest patient with IIH treated with venous sinus stenting being 2 years of age.[14] There were 12 males and 13 females with an average BMI of 27.8 ± 9.8 kg/m². The average BMI of males was 22.78 ± 6.44 kg/m², and the average BMI of females was 36.16 ± 11.98 kg/m², as illustrated in Table 1.

Clinical Presentation

Although the most common presenting symptoms were headache (87.2%), papilledema (51.3%), and vision changes (41.0%), there was a significant degree of heterogeneity in clinical presentation, as depicted in Table 2. The average lumbar puncture opening pressure ± SD was 40.26 ± 11.96 cm H_2O, and on venous manometry the average stenotic pressure ± SD was 13.32 ± 7.7 mm Hg (Table 1). Lee et al. reported a mean lapse from first symptom onset to venous sinus stenting of 5.6 years; however, several cases reported acute intervention.[14,17,19,20] Three cases had no significant past medical history on presentation.[14,16,19] In those that had a history of IIH, acetazolamide was commonly administered.[11,13-16,18,21] The average preoperative disc edema ± SD was 1. 67 ± 1.72 (Frisen scale).

Medical Management

Medical management of pediatric IIH is similar to the treatment of IIH in the adult population. Acetazolamide and loop diuretics such as furosemide are most often prescribed.[22] In the pediatric population, the standard dosage is 25 mg/kg/d; the dose may be increased up to 2 g/d. After this, surgical management is often indicated.[22,23] Additionally, topiramate is used to reduce headache and for its catabolic effects, useful for managing obesity.[24] However, it is important to note that obesity has not been found to be associated with pediatric IIH as opposed to their adult counterparts, and for this reason the applicability of lifestyle interventions is only to those cases with concurrent obesity.[25]

Table 1. Patient Demographics and Diagnostic Cerebrospinal Fluid Characteristics

Study	Number of Patients	Age (years)	M/F	BMI (kg/m²)	LP Opening Pressure	Average Stenotic Pressure (Preoperative)
Boddu et al., 2014[14]	1	2	1/0	21.6	66	14
Rodrigues et al., 2020[20]	1	12	1/0	-	-	11
Carter et al., 2021[16]	1	15	0/1	49.1	55	32
Aguilar-Pérez et al., 2017[11]	4	11.5 ± 5	3/1	28.3 ± 14.0	55 ± 2.5	9.75 ± 4.65
Albuquerque et al., 2011[12]	3	14.0 ± 1.73	1/2	-	-	-
Owler et al., 2003[18]	2	16.5 ± 0.71	0/2	36.0 ± 2.8	32.45 ± 3.6	15.5 ± 13.4
Rajpal et al., 2005[19]	1	15	1/0	26.9	37	25
Barrero Ruiz et al., 2022[13]	1	6	0/1	0	55	23
Lee et al., 2021[17]	14	-	-	24.96 ± 5.7	34.9 ± 8.6	10.8 ± 6.1
Schwarz et al., 2021[21]	7	12.71 ± 5.1	4/3	23.16 ± 6.7	39.0 ± 12.4	14.7 ± 8.7
Buell et al., 2017[15]	1	4	0/1	-	-	23
This study	3	14.3 ± 2.52	0/3	40.58 ± 14.4	35.3 ± 6.3	11.3 ± 4.04
Totals	39	12.28 ± 4.74	11/14	27.8 ± 9.8	40.26 ± 11.96	13.32 ± 7.68

M, male; F, female; BMI, body mass index; LP, lumbar puncture.

Table 2. Presenting Symptoms

Symptom	Preoperative
Headache	34 (87.2)
Papilledema	20 (51.3)
Vision changes	16 (41.0)
Nausea	2 (5.1)
Emesis	5 (12.8)
Tinnitus	4 (10.3)
Diplopia	6 (15.4)
Generalized weakness	3 (7.7)
Altered consciousness	2 (5.1)
Ataxia	1 (2.56)

Values are n (%).

Corticosteroids may be used for short-term management of pediatric IIH, are generally reserved for more severe symptoms, and are often used to manage patients before surgical intervention.[22,24] Lumbar punctures are often used in the management of acutely decompensating IIH, medically refractory papilledema, or as a bridge in managing patients awaiting more invasive surgical intervention.[2,26]

Surgical Management

Aside from venous sinus stenting, there are 2 surgical modalities for managing IIH via CSF diversion: ventricular shunting and optic nerve sheath decompression (ONSD). Ventricular shunting often produces rapid reductions in intracranial pressure and is associated with favorable outcomes.[27] In the past, the lumboperitoneal shunt was popular in the management of pediatric IIH because of the small size of the ventricles causing technical difficulties. However, compared with the ventriculoperitoneal shunt, the lumboperitoneal shunt is associated with worse outcomes. Ventriculoperitoneal shunts have been associated with malfunctions in 3.9% of cases, and lumboperitoneal shunts have been associated with malfunctions in 7.0% of cases ($P < 0.001$).[28] Still, both methods are associated with a significant complication rates, often including overdrainage, shunt malfunction and infection.[28,29] Furthermore, shunt complications have been noted to occur more frequently in younger patients and in patients with a higher BMI.[30] Heyman et al.[31] reported postoperative changes in visual acuities after ventriculoperitoneal shunting and noted that visual deterioration was still seen in 70% of patients secondary to multiple shunt malfunctions within the 4-month follow-up period. Compared with ventriculoperitoneal shunting, dural venous sinus stenting represents a promising minimally invasive alternative, particularly in cases with small ventricle size, and avoids complications associated with shunting including overdrainage, infection, and mechanical malfunctions.

ONSD is specifically used to manage visual symptoms associated with pediatric IIH including visual acuity, visual fields, pupillary light reflex, and papilledema.[32,33] ONSD can effectively improve visual outcomes in pediatric IIH by reducing optic nerve sheath pressure by increasing CSF outflow. A recent study by Bersani et al.[34] compared ONSD in the management of adult and pediatric patients with IIH. They found that children were more likely to present with more severe visual symptoms than their adult counterparts ($P = 0.043$). However, Bersani et al.[34] found that postoperatively there was no significant difference between the visual symptoms of adults and pediatrics ($P = 0.838$). Additionally, a low complication rate was described in both the pediatric and adult populations in association with ONSD. Although good optical outcomes have been described, the efficacy of ONSD in managing IIH is not as robust. Thuente and Buckle[35] documented repeat surgical or medical intervention for refractory IIH after ONSD in 25% of patients ($n = 12$). Consequentially, although ONSD is effective in managing optic symptoms associated with pediatric IIH, close follow-up of intracranial pressure is still warranted in these patients. Additionally, dural venous sinus stenting is also associated with improvement in optic symptoms and represents a valuable alternative, particularly in patients with documented venous sinus stenosis. Unlike ONSD, which primarily targets localized pressure around the optic nerve, dural venous sinus stenting directly addresses the underlying pathophysiology of IIH by restoring venous outflow and reducing global intracranial pressure.

Follow-Up

Of the 39 cases included, the average follow-up $\pm$ SD was 23.0 ± 26.7 months (Table 3). The average change in pre- to postoperative trans-stent pressure $\pm$ SD was -12.1 ± 7.9 mm Hg ($P < 0.001$). Clinical outcome was evaluated as improved, stable, or worse. Improved outcome was defined as significant reduction in all symptoms, stable outcome was defined as no change in one or more symptoms, and worse outcome was defined as worsening of one or more symptoms or acute decompensation necessitating repeat intervention. A total of 21 cases had improved clinical outcome, 2 had stable clinical outcome, and 6 had worse clinical outcome. Ten cases (25.6%) had no available clinical outcomes. Of the patients, 75% ($n = 15$) reported improvement in visual acuity and 73.9% ($n = 17$) reported improvement in papilledema. Of the patients, 5.12% ($n = 2$) had stent-related complications and 15.4% ($n = 6$) had repeat surgery. Postoperative medical management consisted of a combination of acetazolamide and antiplatelet therapy, with medication-related complications occurred in 7.7% ($n = 3$), including epistaxis, menorrhagia, and iron deficiency anemia. The cause of worse clinical outcome was most often an acute episode of IIH after dural venous stenting. There were 6 repeat operations which included one lumbar puncture, one stent placement, and 4 shunt placements.

In this case series, we highlight the efficacy of venous sinus stenting in the management pediatric IIH with evidence of sinus stenosis.[16,17] Furthermore, we also emphasize the challenges associated with postsurgical dual antiplatelet therapy in this population. There are a lack of large trials evaluating the optimal antiplatelet regimen in adults and their pediatric counterparts. Most studies describe dual antiplatelet treatment to consist of doses of aspirin (81–100 mg/d) and clopidogrel (75 mg/d) for 3 to 6 months followed by aspirin (81–100 mg/d) alone from 9 months

Table 3. Dural Venous Sinus Follow-Up Angiographic and Clinical Outcomes

Study	Follow-Up (months)	Average Stent Pressure (Follow-Up)	Change in Pressure Gradient	Repeat Operation	Clinical Outcome, (I/S/W)
Boddu et al., 2014[14]	24	-	-	0	(1/0/0)
Rodrigues et al., 2020[20]	-	1	−10	0	(1/0/0)
Carter et al., 2021[16]	12	6	−26	0	(1/0/0)
Aguilar-Pérez et al., 2017[11]	50.25 ± 24.5	5.3 ± 5.5	−6.3 ± 6.7	0	(4/0/0)
Albuquerque et al., 2011[12]	28 ± 14.14	-	-	0	(1/1/0)
Owler et al., 2003[18]	11.5 ± 0.71	-	-	0	(1/1/0)
Rajpal et al., 2005[19]	6	-	-	0	(1/0/0)
Barrero Ruiz et al., 2022[13]	5	3	−20	1	(0/0/1)
Lee et al., 2021[17]	21.2 ± 37.31	1.67 ± 2.25	−10 ± 7.6	4	(0/0/4)
Schwarz et al., 2021[21]	18.7 ± 11.6	0.86 ± 0.69	−13.9 ± 9.4	0	(7/0/1)
Buell et al., 2017[15]	20	1	−22	0	(1/0/0)
This study	25.15 ± 15.8	2.5 ± 0.71	−11 ± 1.41	1	(3/0/0)
Totals	23.05 ± 26.31	2.34 ± 2.67	−12.1 ± 7.9	6	(21/2/6)

I, improve; S, stable; W, worse.

to a year, which despite large scale trials seems to be sufficient for the adult population.[20,36-39] However, this dose may not be curtailed appropriately when administered to pediatric patients. Further studies are necessary to determine the optimal dosing of postoperative anticoagulants in pediatric populations to reduce the risk of hemorrhagic side effects while still mitigating the risk of thrombosis.

CONCLUSIONS

In this case series, we describe the efficacy of venous angioplasty in the management of pediatric IIH in carefully selected pediatric patients with evidence of venous sinus stenosis. Additionally, we report on the challenges in the postoperative period associated with dual antiplatelet therapy. This study underscores the need for careful patient selection and vigilant postoperative management, and further investigation into optimized antiplatelet regimens that may mitigate these risks without compromising the therapeutic benefits of venous angioplasty in pediatric IIH.

CRediT AUTHORSHIP CONTRIBUTION STATEMENT

Brandon Edelbach: Writing – review & editing, Conceptualization, Formal analysis, Data curation, Writing – original draft. **Fransua Sharafeddin:** Writing – review & editing. **Eman Hawy:** Writing – review & editing. **Timothy Winters:** Writing – review & editing. **Promod Pillai:** Conceptualization, Writing – review & editing, Formal analysis.

REFERENCES

1. Walker RW. Idiopathic intracranial hypertension: any light on the mechanism of the raised pressure? *J Neurol Neurosurg Psychiatry*. 2001;71:1-5.
2. Albakr A, Hamad MH, Alwadei AH, et al. Idiopathic intracranial hypertension in children: diagnostic and management approach. *Sudan J Paediatr*. 2016;16:67-76.
3. Gillson N, Jones C, Reem RE, Rogers DL, Zumberge N, Aylward SC. Incidence and demographics of pediatric intracranial hypertension. *Pediatr Neurol*. 2017;73:42-47.
4. Corbett JJ, Savino PJ, Thompson HS, et al. Visual loss in pseudotumor cerebri. Follow-up of 57 patients from five to 41 years and a profile of 14 patients with permanent severe visual loss. *Arch Neurol*. 1982;39:461-474.
5. Orcutt JC, Page NG, Sanders MD. Factors affecting visual loss in benign intracranial hypertension. *Ophthalmology*. 1984;91:1303-1312.
6. Wall M, George D. Idiopathic intracranial hypertension. A prospective study of 50 patients. *Brain*. 1991;114(pt 1A):155-180.
7. Raoof N, Hoffmann J. Diagnosis and treatment of idiopathic intracranial hypertension. *Cephalalgia*. 2021;41:472-478.
8. Thurtell MJ. Idiopathic intracranial hypertension. *Continuum (Minneap Minn)*. 2019;25:1289-1309.
9. Higgins JNP, Owler BK, Cousins C, Pickard JD. Venous sinus stenting for refractory benign intracranial hypertension. *Lancet*. 2002;359:228-230.
10. Huang LC, Winter TW, Herro AM, et al. Ventriculoperitoneal shunt as a treatment of visual loss in idiopathic intracranial hypertension. *J Neuroophthalmol*. 2014;34:223-228.
11. Aguilar-Pérez M, Martinez-Moreno R, Kurre W, et al. Endovascular treatment of idiopathic intracranial hypertension: retrospective analysis of immediate and long-term results in 51 patients. *Neuroradiology*. 2017;59:277-287.
12. Albuquerque FC, Dashti SR, Hu YC, et al. Intracranial venous sinus stenting for benign intracranial hypertension: clinical indications, technique, and preliminary results. *World Neurosurg*. 2011;75:648-652.
13. Barrero Ruiz E, Iglesias Moroño S, Ros López B, Morales Martinez A, Díaz T, Arráez Sánchez MÁ. Life-threatening idiopathic intracranial hypertension: the role of venous sinus stenting. *Childs Nerv Syst*. 2022;38:1433-1443.

14. Boddu S, Chow G, McConachie N. Successful venous sinus stenting for management of idiopathic intracranial hypertension in a 2 year old: a case report - record details - Embase. 2014. Available at: https://www.embase.com/records?subaction=viewrecord&rid=6&page=2&id=L602325655. Accessed May 1, 2024.

15. Buell TJ, Starke RM, Ding D, et al. Venous sinus stenting using transcranial access for the treatment of idiopathic intracranial hypertension in a pediatric patient. *J Neurosci Rural Pract*. 2017;8:672-675.

16. Carter LM, Chakraborty AR, McCoy-Stephens TM, Strickland AE, Bohnstedt BN, Gross NL. Venous sinus stenosis treatment in pediatric idiopathic intracranial hypertension: illustrative case and literature review. *World Neurosurg*. 2021;149:2-7.

17. Lee KE, Zehri A, Soldozy S, et al. Dural venous sinus stenting for treatment of pediatric idiopathic intracranial hypertension. *J Neurointerv Surg*. 2021;13:465-470.

18. Owler BK, Parker G, Halmagyi GM, et al. Pseudotumor cerebri syndrome: venous sinus obstruction and its treatment with stent placement. *J Neurosurg*. 2003;98:1045-1055.

19. Rajpal S, Niemann DB, Turk AS. Transverse venous sinus stent placement as treatment for benign intracranial hypertension in a young male: case report and review of the literature. *J Neurosurg*. 2005;102:342-346.

20. Rodrigues LHDS, Torres H, Xander P, et al. Cerebral venous sinus stenting for the treatment of idiopathic intracranial hypertension in a child - record details - Embase. 2020. Available at: https://www.embase.com/records?subaction=viewrecord&rid=15&page=1&id=L2020664205. Accessed May 1, 2024.

21. Schwarz J, Al Balushi A, Sundararajan S, et al. Management of idiopathic intracranial hypertension in children utilizing venous sinus stenting. *Interv Neuroradiol*. 2021;27:257-265.

22. Cinciripini GS, Donahue S, Borchert MS. Idiopathic intracranial hypertension in prepubertal pediatric patients: characteristics, treatment, and outcome. *Am J Ophthalmol*. 1999;127:178-182.

23. Honorat R, Marchandot J, Tison C, Cances C, Chaix Y. [Treatment and prognosis of idiopathic intracranial hypertension in children. Retrospective study (1995-2009) and literature review]. *Arch Pediatr*. 2011;18:1139-1147.

24. Jirásková N, Rozsíval P. Idiopathic intracranial hypertension in pediatric patients. *Clin Ophthalmol*. 2008;2:723-726.

25. Aylward SC, Waslo CS, Au JN, Tanne E. Manifestations of pediatric intracranial hypertension from the intracranial hypertension registry. *Pediatr Neurol*. 2016;61:76-82.

26. Spennato P, Ruggiero C, Parlato RS, et al. Pseudotumor cerebri. *Childs Nerv Syst*. 2011;27:215-235.

27. de Melo Junior JO, da Mata Pereira PJ, Niemeyer Filho P. Freehand ventriculoperitoneal shunt for idiopathic intracranial hypertension: technical note for slit-like ventricle puncture. *Cureus*. 2023;15:e34583.

28. Menger RP, Connor DE, Thakur JD, et al. A comparison of lumboperitoneal and ventriculoperitoneal shunting for idiopathic intracranial hypertension: an analysis of economic impact and complications using the Nationwide Inpatient Sample. *Neurosurg Focus*. 2014;37:E4.

29. Sunderland GJ, Jenkinson MD, Conroy EJ, Gamble C, Mallucci CL. Neurosurgical CSF diversion in idiopathic intracranial hypertension: a narrative review. *Life*. 2021;11:393.

30. Galloway L, Karia K, White AM, et al. Cerebrospinal fluid shunting protocol for idiopathic intracranial hypertension for an improved revision rate. *J Neurosurg*. 2021;136:1790-1795.

31. Heyman J, Ved R, Amato-Watkins A, et al. Outcomes of ventriculoperitoneal shunt insertion in the management of idiopathic intracranial hypertension in children. *Childs Nerv Syst*. 2017;33:1309-1315.

32. Hagen K, Ehlis AC, Schneider S, Haeussinger FB, Fallgatter AJ, Metzger FG. Influence of different stimulation parameters on the somatosensory evoked potentials of the nervus vagus–how varied stimulation parameters affect VSEP. *J Clin Neurophysiol*. 2014;31:143-148.

33. Nithyanandam S, Manayath GJ, Battu RR. Optic nerve sheath decompression for visual loss in intracranial hypertension: report from a tertiary care center in South India. *Indian J Ophthalmol*. 2008;56:115-120.

34. Bersani TA, Meeker AR, Sismanis DN, Carruth BP. Pediatric and adult vision restoration after optic nerve sheath decompression for idiopathic intracranial hypertension. *Orbit*. 2016;35:132-139.

35. Thuente DD, Buckley EG. Pediatric optic nerve sheath decompression. *Ophthalmology*. 2005;112:724-727.

36. Gurney SP, Ramalingam S, Thomas A, Sinclair AJ, Mollan SP. Exploring the current management idiopathic intracranial hypertension, and understanding the role of dural venous sinus stenting. *Eye Brain*. 2020;12:1-13.

37. Reid K, Winters HS, Ang T, Parker GD, Halmagyi GM. Transverse sinus stenting reverses medically refractory idiopathic intracranial hypertension. *Front Ophthalmol*. 2022;2:885583.

38. Sheriff F, Inam ME, Thanh Truong VT, et al. Dual antiplatelet therapy duration after venous sinus stenting for idiopathic intracranial hypertension and stent survival-is longer necessarily better? A meta-regression. *World Neurosurg*. 2021;151:e86-e93.

39. Teleb MS, Cziep ME, Lazzaro MA, et al. Idiopathic intracranial hypertension: a systematic analysis of transverse sinus stenting. *Interv Neurol*. 2014;2:132-143.

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